

- 1 The radiancy R is defined as the total energy emitted per unit time per unit area from a black body at thermodynamic temperature T . It is given by

$$R = \sigma T^4$$

where σ is called the Stefan-Boltzmann constant.

What is the correct unit for σ ?

- A $\text{W m}^{-2} \text{K}^4$
- B $\text{kg s}^{-3} \text{K}^{-4}$
- C $\text{J s m}^{-2} \text{K}^{-4}$
- D $\text{J s}^{-1} \text{m}^{-2} \text{K}^4$